NEBIUS × NVIDIA HACKATHON · APPS AND AGENTS

CRISPR-Ops

Don't just score the guide. Audit the assumption.

An autonomous audit of CRISPR off-target scores when the reference genome is not the allele a named population actually carries.

Team: Andrew Benjamin, Will, Mike

Research use only. Not a diagnosis. Personal genomes are refused.

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The reference genome is an assumption

Guide-RNA design tools score a spacer against one reference sequence and a mismatch count. Both can be the wrong object. A common variant can leave the mismatch count unchanged and still change the CFD score, because the mismatched base pair is now a different one. A variant can also land on a base the score ignores, such as the N of an NGG PAM, and the number does not move at all.

CRISPR-Ops takes a gene, a gnomAD population, and a panel of guide candidates. It audits how off-target scores behave under that population's common variants. The output is a method result with a trace of every tool call. It is not a clinical screen.

INPUT

Gene, population, 20 nt spacers. An optional note is checked against the refusal policy before any tool runs.

LOOP

Plan, act, verify, revise. The next tool depends on the last observation.

THRESHOLD

$\tau = 0.2$ CFD, registered before the run. Common variant means allele frequency ≥ 0.05.

SCOPE

Substitutions only. GRCh38 chromosome 11 in the current demo. Bulges and indels are out of scope.

What a verdict means

A guide is FLAGGED when any non-on-target site has CFD ≥ 0.2, and CLEAR otherwise. A flip is a change between the reference haplotype and the population haplotype on the NGG wave. A score that stays below 0.2 is a statement about the scoring method, not a statement that a site is safe.

An orchestrator with deterministic hands

The model is not a writer appended to a finished pipeline. When `NVIDIA_API_KEY` is set, Nemotron (`nvidia/nvidia-nemotron-nano-9b-v2`) chooses the next tool through the NeMo Agent Toolkit ReAct workflow. It may choose only from the legal set for that step. A choice outside the set is rejected and written into the trace. A sentence that adds a number the tools did not return is dropped.

Without the key, the model is not called. The registered policy chooses the branch, and the first line of the trace says so. A flipped verdict remains a consequence of $\tau = 0.2$. If the model says to hold a PAM expansion after a flip, the policy keeps the expansion and records the override.

Decisions that make it an agent

DECISION	WHAT IT DOES
Triage	Evo 2 is spent only on sites with $\lvert\Delta CFD\rvert \geq 0.05$ or a score within 0.05 of 0.2, capped at four sites. The orchestrator may defer that spend.
Disagreement	When the engine and the second checker disagree, the orchestrator names a hypothesis: strand, window start, or phasing. All three are tested. The trace says which one the tests support.
Escalation	NAG and NGA are searched for one guide only, after its NGG verdict flips or a corrected checker still disagrees. Other guides stay on NGG.
Refusal	A personal genome or a clinical question is rejected before enumeration. The refusal is logged.

Tools

TOOL	CONTRACT
enumerate_sites	Guide, PAM set, mismatch ≤ 4. CPU enumerator. Both strands. 23 bp window, mismatches on the 20 bp spacer.
fetch_af	gnomAD v4 allele frequency for one population. SNVs only. Disk cache. Alleles that contradict the FASTA are dropped.
score_cfd	Doench 2016 CFD on the reference window and the alternate window. Position 1 is PAM-distal. The PAM N is not in the product.
crispr_me_check	Second enumerator plus a second window builder. In-repo unless CRISPRME_BIN is set. It is not CRISPRme+ by default.
resolve_disagreement	Compares haplotype deltas. Investigate, expand PAM, ask Evo 2, or accept. Accept is not a safety verdict.
evo2_score	Evo 2 NIM mean log-probability of the reference window versus the variant window. A sequence-context prior, not a cutting probability.
gpu_enumerate	Same hit definition as the CPU scanner, in cuda/enumerate.cu. Used for the expanded PAM search when the binary exists.
verify_citation	A claim is shown only if its numbers are in the run record and it is not a clinical conclusion.

The GPU is a decision, and the null is allowed

AGENCY

Triage, disagreement, escalation, and refusal are branches. The trace is the product, not a summary glued on at the end.

TWO SCORING AXES

CFD is a lookup. Evo 2 is the second axis, used where CFD moves or sits near τ. Likelihood is a prior, never ground truth.

ADAPTIVE PAM SCOPE

NAG and NGA are not brute-forced for every guide. The search widens only for the guide whose verdict moved.

VISIBLE CORRECTION

A real strand bug in the second window builder is detected, hypothesized, re-run, and closed in the trace.

The scientific bar is the part that should not be softened for a demo. CFD is not treated as a validated oracle. A GUIDE-seq table is scored only if `data/guideseq.tsv` is present. If population-aware ranking does not beat the reference-only ranking, the result is the null. This build does not claim a correlation.

Absolute CFD values are reported against the registered threshold. On the chromosome 11 panel, the one common South Asian overlap does not move CFD. That is reported as method behavior: the substitution is the N of the PAM, and CFD multiplies mismatch weights and the last two PAM bases.

The demo does not fill in an allele frequency, a CFD split, or an experimental correlation that this run did not measure. The reference base at the overlapping site is not a minority allele.

Loop, tools, and the GPU layer

User
Gene, population, guide panel, optional note.

→

Orchestrator
NeMo Agent Toolkit `react_agent`. Nemotron when a key is set. Registered policy otherwise. Policy refusal runs first.

→

Trace and report
Tool calls, hypothesis, re-run, citations, limitations.

DETERMINISTIC TOOLS
enumerate_sites, fetch_af, score_cfd, crispr_me_check, resolve_disagreement, verify_citation. Pure functions. They do not choose the next step.

GPU LAYER
evo2_score on the BioNeMo NIM. gpu_enumerate from cuda/enumerate.cu. Called when triage or escalation says the cheap pass is not enough.

Enumeration

The CPU binary in `cpp/enumerate.cpp` has two algorithms, PAM-skip and a full scan. Their hit lists are required to match. The CUDA kernel uses the same hit definition and the same TSV. On this machine there is no `nvcc` binary, so the expanded search falls back to the CPU enumerator and the trace says which device ran. A measured chromosome 11 NGG scan of one guide was about 1.3 seconds for PAM-skip and 1.5 seconds for the full scan, with equal hit lists.

Haplotype rule

The primary haplotype is the single highest-frequency SNV in the 23 bp window. gnomAD does not phase that window. A compound all-in-cis haplotype is recorded as a sensitivity when the checker builds one. It is not adopted as the verdict.

NeMo Agent Toolkit

`workflows/crispr_ops.yml` is the ReAct workflow: ten tools, `max_tool_calls: 40`, native tool calling, model `nvidia/nvidia-nemotron-nano-9b-v2`. `nat validate` accepts it. `workflows/popoff_audit.yml` runs the same audit with the policy choosing every branch, which is the path the local demo uses until a key is set.

HBB, South Asian, chromosome 11

Three guides. HBB-HTY3 is a published exon-1 spacer against the reference allele. HBB-EX1-A and HBB-PROM are NGG sites read off GRCh38 at this locus. They are audit candidates, not therapeutic recommendations.

GUIDE	REFERENCE	POPULATION	FLIP
HBB-HTY3	CLEAR	CLEAR	No
HBB-EX1-A	FLAGGED	FLAGGED	No
HBB-PROM	FLAGGED	FLAGGED	No

Nineteen off-target sites were enumerated. One common South Asian SNV overlaps a window: chr11:13061395 C>T, allele frequency 0.25, on HBB-PROM at window chr11:13061375. CFD stays 0.0391 because the base is the N of the PAM. No guide flipped, so NAG and NGA were not searched. The second checker disagreed on minus-strand windows. The policy hypothesis was strand. The re-run agreed. The resolver moved from INVESTIGATE to ACCEPT.

Evo 2 was not scored, because the NVIDIA API key is unset and the endpoint was not called. The CUDA kernel was not timed.

Two-minute demo, from this run

TIME	WHAT IS ON SCREEN
0:00–0:20	The reference allele and the population allele are not the same object. At this site the population base is common and CFD does not move, because it is the N of the PAM.
0:20–1:00	Type HBB and South Asian. The trace streams tool calls, a strand disagreement, and the re-run that agrees. NAG expansion does not fire, and the trace says why.
1:00–1:30	One site: ref and alt windows, CFD 0.0391 to 0.0391, the PAM-N reading. Evo 2 is deferred and labeled unavailable.
1:30–2:00	CPU PAM-skip versus full scan, equal hits, no GPU timing, no GUIDE-seq table, the limitations list.

Limitations, stated with the result

- The loaded reference is GRCh38 chromosome 11, not the whole genome.
- Substitutions only. The primary haplotype is one SNV, because gnomAD does not phase the window.
- The cross-check is an in-repo second enumerator unless CRISPRME_BIN points at an external binary.
- $\tau = 0.2$ is a registered method cutoff, not a measured cutting rate.
- Population-by-guide-by-site scoring across every gnomAD superpopulation was not built. This build does the enumeration kernel and the Evo 2 prior.

Research-use only. This is an audit of how off-target scores behave under population variation. It is not a clinical screen, not a diagnosis, and it does not accept personal genomes. Evo 2 likelihood is a sequence-context prior, not a cutting probability.

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